Telco Churn Analysis Project

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

import warnings
warnings.filterwarnings('ignore')

df = pd.read_csv('Customer Churn.csv')

pd.set_option('display.max_columns',None)
df.head(2)
```

```
   customerID  gender  SeniorCitizen Partner Dependents  tenure
PhoneService  \
0  7590-VHVEG  Female              0     Yes         No       1
No
1  5575-GNVDE    Male              0      No         No      34
Yes

      MultipleLines InternetService OnlineSecurity OnlineBackup  \
0  No phone service             DSL             No          Yes
1                No             DSL            Yes           No

  DeviceProtection TechSupport StreamingTV StreamingMovies
Contract  \
0               No          No          No              No  Month-to-
month
1              Yes          No          No              No        One
year

  PaperlessBilling     PaymentMethod  MonthlyCharges TotalCharges
Churn
0              Yes  Electronic check           29.85        29.85
No
1               No      Mailed check           56.95       1889.5
No
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
```

```
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'],
errors='coerce')
```

```
df['TotalCharges'].isnull().sum()
```

```
np.int64(11)
```

```
df['TotalCharges'].fillna(0,inplace=True)
```

```
df['TotalCharges'].isnull().sum()
```

```
np.int64(0)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
```

```
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   float64
 20  Churn             7043 non-null   object
dtypes: float64(2), int64(2), object(17)
memory usage: 1.1+ MB
```

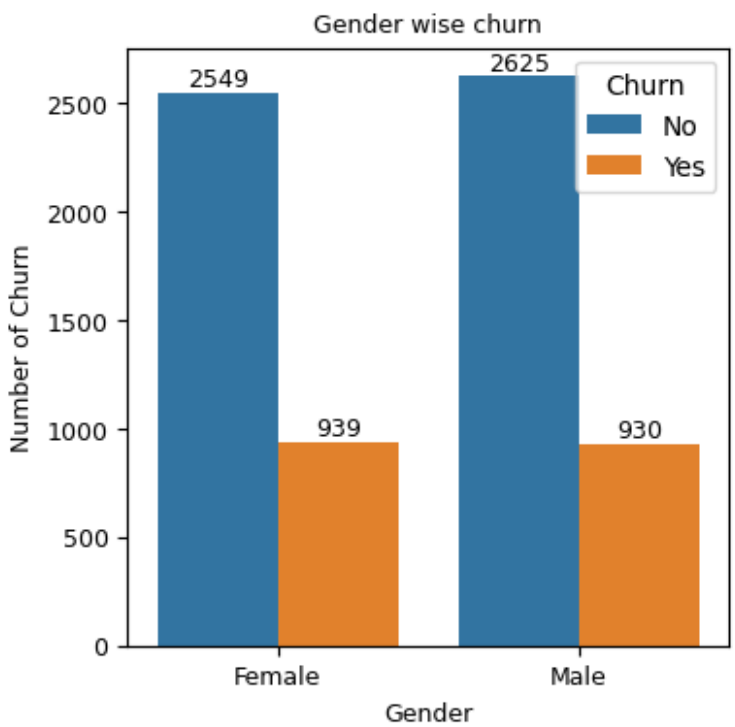
```
df.head(2)
```

```
   customerID  gender  SeniorCitizen Partner Dependents  tenure
PhoneService  \
0  7590-VHVEG  Female              0     Yes         No       1
No
1  5575-GNVDE    Male              0      No         No      34
Yes

      MultipleLines InternetService OnlineSecurity OnlineBackup  \
0  No phone service             DSL             No          Yes
1                No             DSL            Yes           No

  DeviceProtection TechSupport StreamingTV StreamingMovies
Contract  \
0               No          No          No              No  Month-to-
month
1              Yes          No          No              No        One
year

  PaperlessBilling     PaymentMethod  MonthlyCharges  TotalCharges
Churn
0              Yes  Electronic check           29.85         29.85
No
1               No      Mailed check           56.95       1889.50
No
```

```
gender_wise_churn =
df.groupby(['gender','Churn'],as_index=False).size()
gender_wise_churn
```

```
   gender Churn  size
0  Female    No  2549
1  Female   Yes   939
2    Male    No  2625
3    Male   Yes   930
```

```
plt.figure(figsize=(4,4))
ax = sns.barplot(data=gender_wise_churn,
x='gender',y='size',hue='Churn')

for bars in ax.containers:
    ax.bar_label(bars, fontsize=9)

ax.tick_params(axis='x',labelsize=9)
ax.tick_params(axis='y',labelsize=9)
plt.title('Gender wise churn',fontsize=9)
plt.ylabel("Number of Churn",fontsize=9)
plt.xlabel("Gender",fontsize=9)
plt.show()
```



#Churn distribution is almost identical across genders, indicating that gender does not significantly impact customer churn.

```
SCE = {0:"Not Senior Citizen",1:"Senior Citizen"}
df['SCEncoding'] = df['SeniorCitizen'].map(SCE)
```

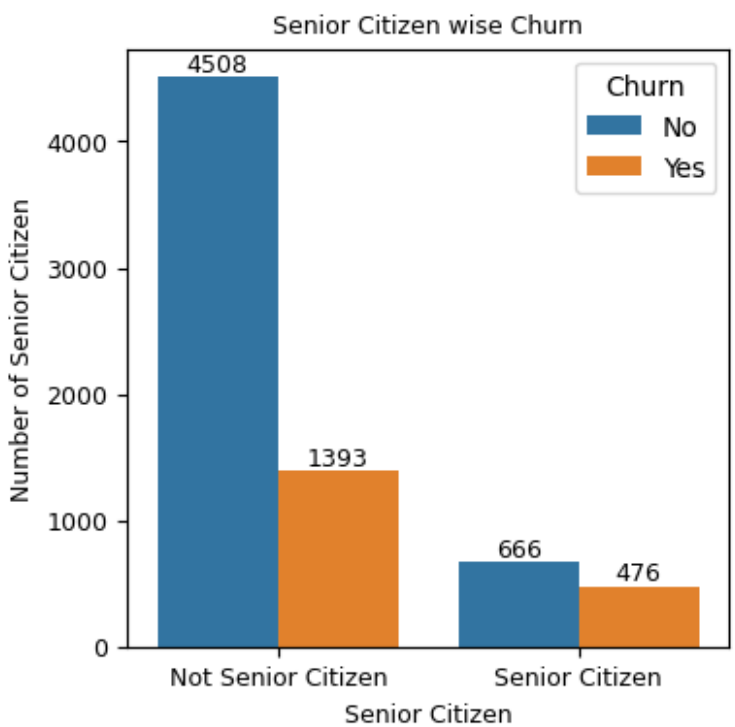
```
df = df[['customerID', 'gender', 'SeniorCitizen', 'SCEncoding',
'Partner', 'Dependents',
       'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
       'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract',
       'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges',
       'TotalCharges', 'Churn']]
df.head(2)
```

```
   customerID  gender  SeniorCitizen          SCEncoding Partner
Dependents  \
0  7590-VHVEG  Female              0  Not Senior Citizen     Yes
No
1  5575-GNVDE    Male              0  Not Senior Citizen      No
No

   tenure PhoneService     MultipleLines InternetService
OnlineSecurity  \
0       1           No  No phone service             DSL
No
1      34          Yes                No             DSL
Yes

  OnlineBackup DeviceProtection TechSupport StreamingTV
StreamingMovies  \
0          Yes               No          No          No
No
1           No              Yes          No          No
No

         Contract PaperlessBilling     PaymentMethod
MonthlyCharges  \
0  Month-to-month              Yes  Electronic check           29.85

1        One year               No      Mailed check           56.95


   TotalCharges Churn
0         29.85    No
1       1889.50    No
```

```
SC_wise_churn =
df.groupby(['SCEncoding','Churn'],as_index=False).size()
SC_wise_churn
```

```
           SCEncoding Churn  size
0  Not Senior Citizen    No  4508
1  Not Senior Citizen   Yes  1393
2      Senior Citizen    No   666
3      Senior Citizen   Yes   476
```

```
plt.figure(figsize=(4,4))
ax = sns.barplot(data=SC_wise_churn, x='SCEncoding',y='size',
hue='Churn')

for bars in ax.containers:
    ax.bar_label(bars, fontsize=9)

ax.tick_params(axis='x',labelsize=9)
ax.tick_params(axis='y',labelsize=9)


plt.title('Senior Citizen wise Churn',fontsize=9)
plt.ylabel('Number of Senior Citizen',fontsize=9)
plt.xlabel('Senior Citizen',fontsize=9)
plt.show()
```



#Senior citizens exhibit a noticeably higher churn rate compared to non-senior customers, despite representing a smaller portion of the total customer base.

```
df.head(2)
```

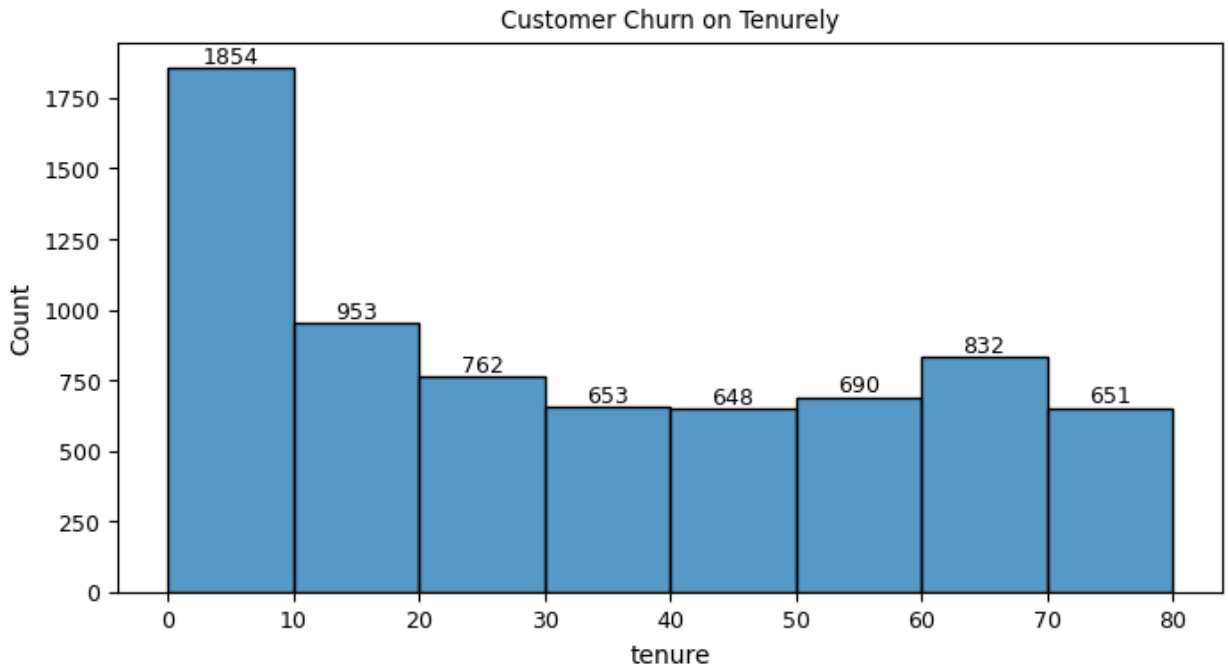
```
   customerID  gender  SeniorCitizen          SCEncoding Partner
Dependents  \
0  7590-VHVEG  Female              0  Not Senior Citizen     Yes
No
1  5575-GNVDE    Male              0  Not Senior Citizen      No
No
```

```
   tenure PhoneService     MultipleLines InternetService
OnlineSecurity  \
0       1           No  No phone service             DSL
No
1      34          Yes                No             DSL
Yes

  OnlineBackup DeviceProtection TechSupport StreamingTV
StreamingMovies  \
0          Yes               No          No          No
No
1           No              Yes          No          No
No

         Contract PaperlessBilling     PaymentMethod
MonthlyCharges  \
0  Month-to-month              Yes  Electronic check           29.85

1        One year               No      Mailed check           56.95


   TotalCharges Churn
0         29.85    No
1       1889.50    No
```

```
(df['tenure'].min(),df['tenure'].max())
```

```
(0, 72)
```

```
plt.figure(figsize=(8,4))
ax = sns.histplot(data=df['tenure'],bins=[0,10,20,30,40,50,60,70,80])

for bars in ax.containers:
    ax.bar_label(bars,fontsize=9)

ax.tick_params(axis='x',labelsize=9)
ax.tick_params(axis='y',labelsize=9)
plt.title('Customer Churn on Tenurely',fontsize=9.5)
plt.show()
```



#Customer churn is significantly higher among customers with lower tenure, indicating that new customers are more likely to leave compared to long-term customers.

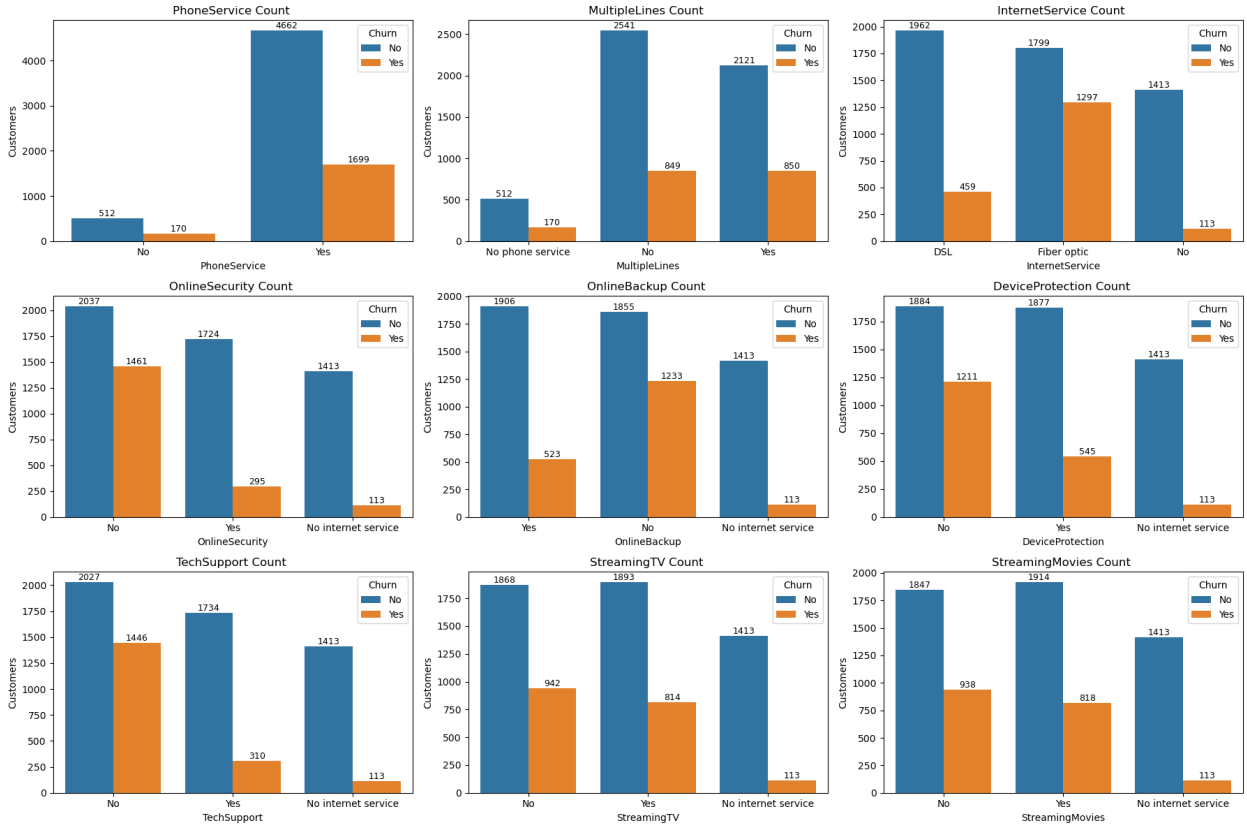
```
df.head(2)

   customerID  gender  SeniorCitizen          SCEncoding Partner
Dependents  \
0  7590-VHVEG  Female              0  Not Senior Citizen     Yes
No
1  5575-GNVDE    Male              0  Not Senior Citizen      No
No

   tenure PhoneService     MultipleLines InternetService
OnlineSecurity  \
0       1           No  No phone service             DSL
No
1      34          Yes                No             DSL
Yes

  OnlineBackup DeviceProtection TechSupport StreamingTV
StreamingMovies  \
0          Yes               No          No          No
No
1           No              Yes          No          No
No

         Contract PaperlessBilling     PaymentMethod
MonthlyCharges  \
0  Month-to-month              Yes  Electronic check           29.85
```

```
1        One year                No      Mailed check           56.95


   TotalCharges Churn
0         29.85    No
1       1889.50    No
```

```
df.columns.values
```

```
array(['customerID', 'gender', 'SeniorCitizen', 'SCEncoding',
'Partner',
       'Dependents', 'tenure', 'PhoneService', 'MultipleLines',
       'InternetService', 'OnlineSecurity', 'OnlineBackup',
       'DeviceProtection', 'TechSupport', 'StreamingTV',
       'StreamingMovies', 'Contract', 'PaperlessBilling',
'PaymentMethod',
       'MonthlyCharges', 'TotalCharges', 'Churn'], dtype=object)
```

```
services = ['PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
       'TechSupport', 'StreamingTV', 'StreamingMovies']

fig,axes = plt.subplots(3,3,figsize=(18,12))

axes = axes.flatten()

for idx, col in enumerate(services):
    ax = sns.countplot(data=df, x=col, ax=axes[idx],hue='Churn')

    for bars in ax.containers:
        ax.bar_label(bars,fontsize=(9))
    ax.set_title(f"{col} Count ")
    ax.set_ylabel("Customers")

plt.tight_layout()
plt.show()
```



#Customers without Online Security have significantly higher churn compared to those who have it.

#Lack of Tech Support is strongly associated with increased churn.

#Customers without Online Backup also show higher churn behavior.

#Fiber optic Internet users have a noticeably higher churn rate compared to DSL users.

#Services like PhoneService and MultipleLines show relatively less impact on churn.

#Customers with No Internet Service generally have lower churn counts.

#Value-added services (Security, Support, Backup) appear to play a crucial role in customer retention.

```
df.head(2)
```

```
   customerID  gender  SeniorCitizen         SCEncoding Partner
Dependents  \
0  7590-VHVEG  Female              0  Not Senior Citizen     Yes
No
1  5575-GNVDE    Male              0  Not Senior Citizen      No
No

   tenure PhoneService     MultipleLines InternetService
```

```
OnlineSecurity  \
0       1           No  No phone service             DSL
No
1      34          Yes                No             DSL
Yes

  OnlineBackup DeviceProtection TechSupport StreamingTV
StreamingMovies  \
0          Yes               No          No          No
No
1           No              Yes          No          No
No

         Contract PaperlessBilling     PaymentMethod
MonthlyCharges  \
0  Month-to-month              Yes  Electronic check           29.85

1        One year               No      Mailed check           56.95


   TotalCharges Churn
0         29.85    No
1       1889.50    No
```

```
Contract_wise_churn =
df.groupby(['Contract','Churn']).size().unstack()
Contract_wise_churn
```

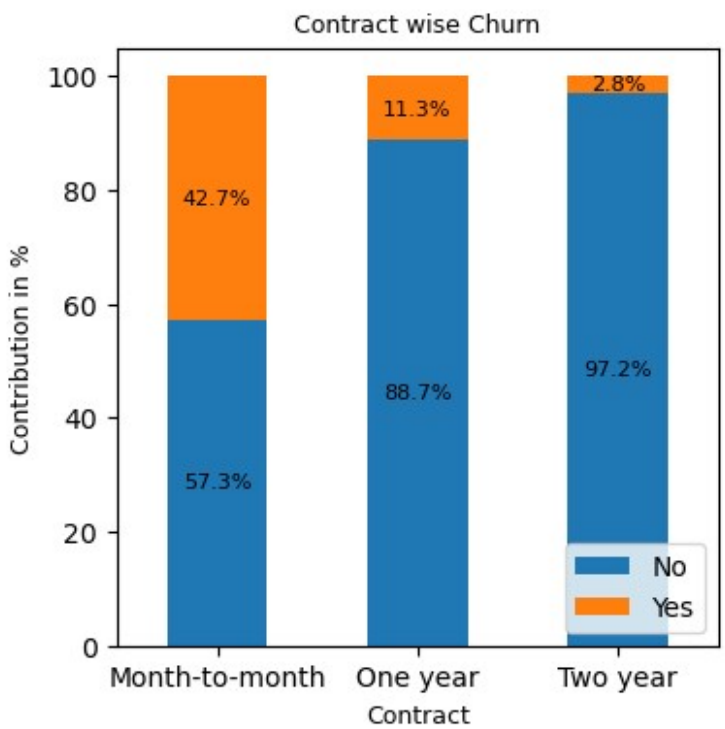
```
Churn             No   Yes
Contract
Month-to-month  2220  1655
One year        1307   166
Two year        1647    48
```

```
Contract_wise_churn_per =
Contract_wise_churn.div(Contract_wise_churn.sum(axis=1),axis=0) * 100
Contract_wise_churn_per
```

```
Churn                  No        Yes
Contract
Month-to-month  57.290323  42.709677
One year        88.730482  11.269518
Two year        97.168142   2.831858
```

```
ax =
Contract_wise_churn_per.plot(kind='bar',stacked=True,figsize=(4,4))

for idx, contract in enumerate(Contract_wise_churn_per.index):
    total=0
    for status in Contract_wise_churn_per.columns:
        value = Contract_wise_churn_per.loc[contract, status]
```

```
        if value > 0:
            ax.text(idx, total+(value/2), f"{value:.1f}%",
ha='center', va='center',fontsize=8)

        total+=value

ax.tick_params(axis='x',rotation=0)
plt.legend(loc='lower right')

plt.title('Contract wise Churn',fontsize=9)
plt.ylabel('Contribution in %',fontsize=9)
plt.xlabel('Contract',fontsize=9)
plt.show()
```



#Customers on month-to-month contracts have the highest churn rate (42.7%), indicating lower retention and potential dissatisfaction or lack of commitment.

#As contract length increases, churn drops significantly: 11.3% for one-year contracts and only 2.8% for two-year contracts.

#The majority of customers in longer contracts (one-year and two-year) are retained (88.7% and 97.2% respectively).

```
df.head(2)
```

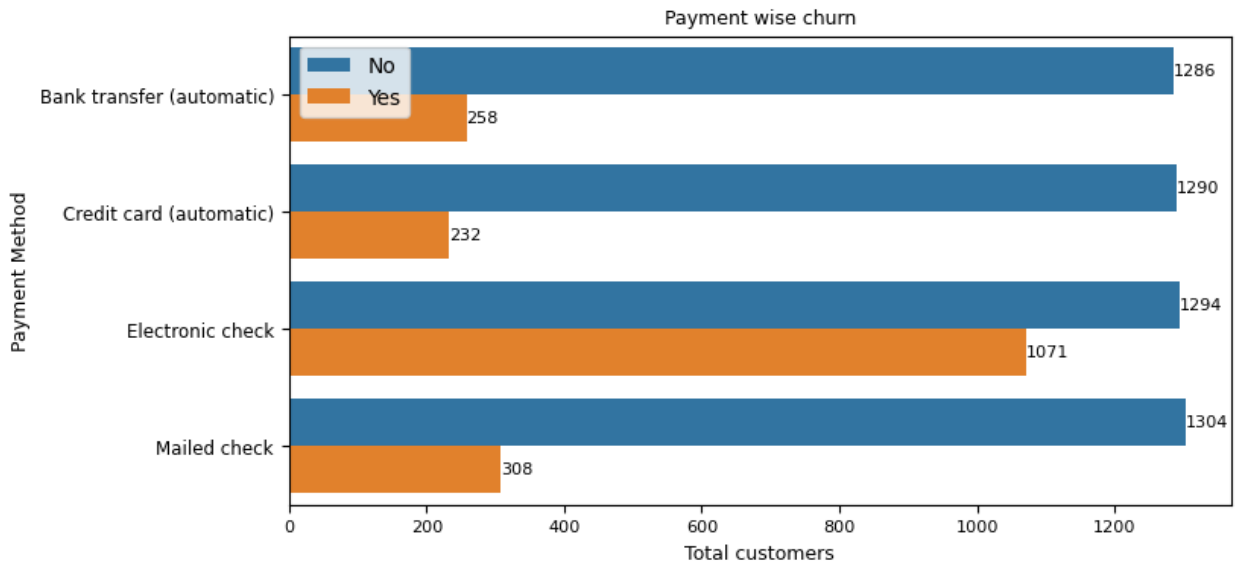
```
   customerID  gender  SeniorCitizen          SCEncoding Partner
Dependents  \
0  7590-VHVEG  Female              0  Not Senior Citizen     Yes
No
1  5575-GNVDE    Male              0  Not Senior Citizen      No
No

   tenure PhoneService     MultipleLines InternetService
OnlineSecurity  \
0       1           No  No phone service             DSL
No
1      34          Yes                No             DSL
Yes

  OnlineBackup DeviceProtection TechSupport StreamingTV
StreamingMovies  \
0          Yes               No          No          No
No
1           No              Yes          No          No
No

         Contract PaperlessBilling     PaymentMethod
MonthlyCharges  \
0  Month-to-month              Yes  Electronic check           29.85

1        One year               No      Mailed check           56.95


   TotalCharges Churn
0         29.85    No
1       1889.50    No
```

```
PaymentMethod_wise_churn =
df.groupby(['PaymentMethod','Churn'],as_index=False).size()
PaymentMethod_wise_churn
```

```
               PaymentMethod Churn  size
0  Bank transfer (automatic)    No  1286
1  Bank transfer (automatic)   Yes   258
2    Credit card (automatic)    No  1290
3    Credit card (automatic)   Yes   232
4           Electronic check    No  1294
5           Electronic check   Yes  1071
6               Mailed check    No  1304
7               Mailed check   Yes   308
```

```
plt.figure(figsize=(8,4))
ax = sns.barplot(data=PaymentMethod_wise_churn,
y='PaymentMethod',x='size',hue='Churn')

for bars in ax.containers:
```

```
    ax.bar_label(bars, fontsize=8)

ax.tick_params(axis='x',labelsize=8)
ax.tick_params(axis='y',labelsize=8.5)

plt.title('Payment wise churn',fontsize=9)
plt.ylabel("Payment Method",fontsize=9)
plt.xlabel("Total customers",fontsize=9)
plt.legend(loc='upper left')
plt.show()
```

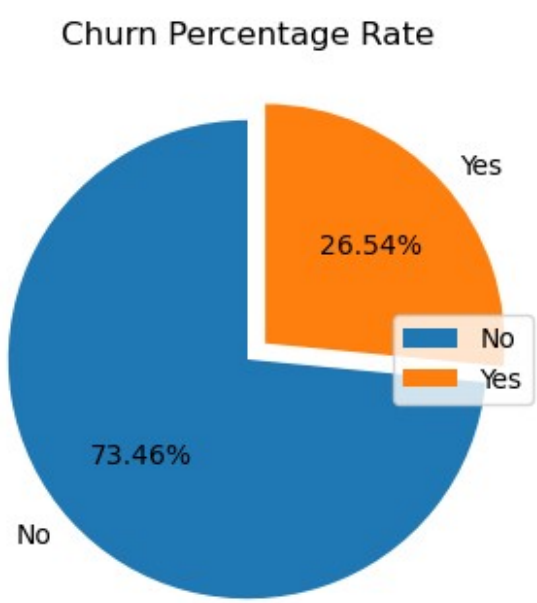


#Customers using (electronic checks and Mailed checks) to show the highest churn rate at 80.9%, while those with automatic payment methods like (bank transfer and credit card) are far more likely to stay, with churn rates below 20%.

```
df.head(2)

   customerID  gender  SeniorCitizen          SCEncoding Partner
Dependents  \
0  7590-VHVEG  Female              0  Not Senior Citizen     Yes
No
1  5575-GNVDE    Male              0  Not Senior Citizen      No
No

   tenure PhoneService     MultipleLines InternetService
OnlineSecurity  \
0       1           No  No phone service             DSL
No
1      34          Yes                No             DSL
Yes

  OnlineBackup DeviceProtection TechSupport StreamingTV
```

```
StreamingMovies  \
0          Yes                No           No           No
No
1           No               Yes           No           No
No

         Contract PaperlessBilling     PaymentMethod
MonthlyCharges  \
0  Month-to-month              Yes  Electronic check           29.85

1        One year               No      Mailed check           56.95


   TotalCharges Churn
0         29.85    No
1       1889.50    No
```

```
exp = [0,0.1]
churnCount = df['Churn'].value_counts(normalize=True) * 100

churnCount.plot(kind='pie',autopct="%1.2f%%", explode=exp,
startangle=90, figsize=(4,4))

plt.title('Churn Percentage Rate')
plt.ylabel('')
plt.legend(loc='center right')
plt.show()
```



#26.54% of customers have churned, while the majority (73.46%) remain active.This indicates that roughly one highlighting a significant retention challenge.